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Inhalt der Vorlesung 1/2

Kapitel

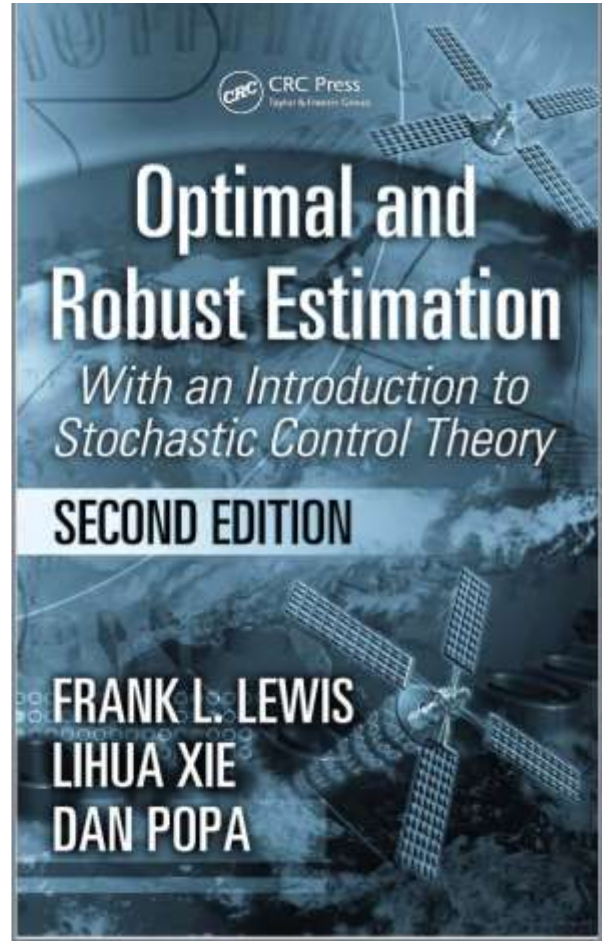
- **Kapitel 1: Übersicht über Unmanned Aerial Systems (UAVs) (1 W)**
- **Kapitel 2: Wesentliche Komponenten/Subsysteme eines UAV (1 W)**
- **Kapitel 3: Modellierung und Simulation von UAVs (1 W)**
- **Kapitel 4: Flight State Controller (PD Cascade) (1W)**
- **Kapitel 5: Pfadplanung (4 W)**
- **Kapitel 6: Perception und Sensing (1 W)**
- **Kapitel 7: Navigation (2 W)**
- **Kapitel 8: Multi-Vehicle Cooperation and Coordination (1 W)**

Reference

Frank L. Lewis (Life Fellow, IEEE) received the bachelor's degree in physics/electrical engineering and the M.S.E.E. degree from Rice University, Houston, TX, USA, in 1971, the M.S. degree in aeronautical engineering from the University of Florida, Gainesville, FL, USA, in 1977, and the Ph.D. degree from Georgia Tech, Atlanta, GA, USA, in 1988.

He is a UTA Charter Distinguished Scholar Professor, a UTA Distinguished Teaching Professor, and the Moncrief-O'Donnell Chair with Arlington Research Institute, University of Texas at Arlington, Arlington, TX, USA. He ranked at position 88 worldwide, 66 in the USA, and three in Texas of all scientists in computer science and electronics, by Guide2Research.com. He has 80 000 Google Scholar Citations. He works in feedback control, intelligent systems, reinforcement learning, cooperative control systems, and nonlinear systems. He is the author of seven U.S. patents, numerous journal special issues, 445 journal articles, 20 books, including the textbooks *Optimal Control*, *Aircraft Control*, *Optimal Estimation*, and *Robot Manipulator Control*.

Dr. Lewis received the Fulbright Research Award, the NSF Research Initiation Grant, the ASEE Terman Award, the International Neural Network Society Gabor Award, the U.K. Institute of Measurement and Control Honeywell Field Engineering Medal, the IEEE Computational Intelligence Society Neural Networks Pioneer Award, the AIAA Intelligent Systems Award, and the AACC Ragazzini Award. He is a Fellow of the National Academy of Inventors, IFAC, AAAS, the U.K. Institute of Measurement and Control, and PE Texas; and U.K. Chartered Engineer.



[Optimal and Robust Estimation - Google Books]

Estimators / Observer

- **Problem:** So far we have assumed that we have full access to the state $\mathbf{x}(t)$ when we designed our controllers.
  - Most often all of this information is not available.
  - And certainly there is usually error in our knowledge of $\mathbf{x}$.

- Usually can only feedback information that is developed from the sensors measurements.
  - Could try "output feedback" $\mathbf{u} = K\mathbf{x} \quad \Rightarrow \quad \mathbf{u} = \hat{K}\mathbf{y}$
  - But this is type of controller is hard to design.

- **Alternative approach:** Develop a replica of the dynamic system that provides an "estimate" of the system states based on the measured output of the system.

- **New plan:** called a "separation principle"
  1. Develop estimate of $\mathbf{x}(t)$, called $\hat{\mathbf{x}}(t)$.
  2. Then switch from $\mathbf{u} = -K\mathbf{x}(t)$ to $\mathbf{u} = -K\hat{\mathbf{x}}(t)$.

- Two key questions:
  - How do we find $\hat{\mathbf{x}}(t)$?
  - Will this new plan work? (yes, and very well)

Estimation Schemes

- Assume that the system model is of the form:

$$\begin{aligned} \dot{\mathbf{x}} &= A\mathbf{x} + B\mathbf{u}\,, \quad \mathbf{x}(0) \text{ unknown} \\ \mathbf{y} &= C_y\mathbf{x} \end{aligned}$$

  where
  - A, B, and C_y are known – possibly time-varying, but that is suppressed here.
  - $\mathbf{u}(t)$ is known
  - Measurable outputs are $\mathbf{y}(t)$ from $C_y \neq I$

- **Goal:** Develop a dynamic system whose state

$$\hat{\mathbf{x}}(t) = \mathbf{x}(t) \quad \forall t \geq 0$$

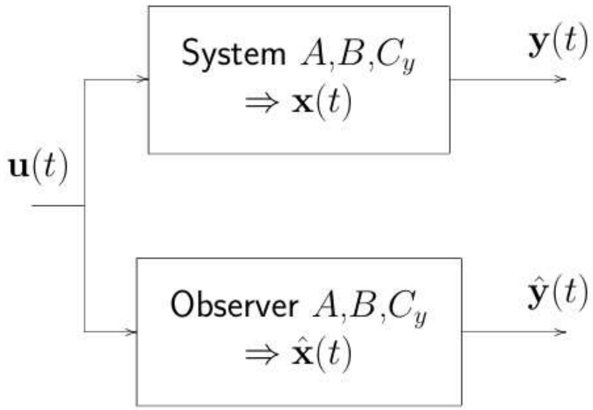
  Two primary approaches:
  - Open-loop.
  - Closed-loop.

Open-loop Estimator 1

- Given that we know the plant matrices and the inputs, we can just perform a simulation that runs in parallel with the system

$$\dot{\hat{\mathbf{x}}}(t) = A\hat{\mathbf{x}} + B\mathbf{u}(t)$$

  – Then $\hat{\mathbf{x}}(t) \equiv \mathbf{x}(t)\ \forall\ t$ provided that $\hat{\mathbf{x}}(0) = \mathbf{x}(0)$



- To analyze this case, start with:

$$\begin{aligned}\dot{\mathbf{x}}(t) &= A\mathbf{x}(t) + B\mathbf{u}(t)\\ \dot{\hat{\mathbf{x}}}(t) &= A\hat{\mathbf{x}}(t) + B\mathbf{u}(t)\end{aligned}$$

- Define the **estimation error**: $\tilde{\mathbf{x}}(t) = \mathbf{x}(t) - \hat{\mathbf{x}}(t)$.
  – Now want $\tilde{\mathbf{x}}(t) = 0\ \forall\ t$, but is this realistic?

- **Major Problem:** We do not know $\mathbf{x}(0)$

Open-loop Estimator 2

- Subtract to get:

$$\frac{d}{dt}(\mathbf{x} - \hat{\mathbf{x}}) = A(\mathbf{x} - \hat{\mathbf{x}}) \quad \Rightarrow \quad \dot{\tilde{\mathbf{x}}}(t) = A\tilde{\mathbf{x}}$$

  which has the solution

$$\tilde{\mathbf{x}}(t) = e^{At}\tilde{\mathbf{x}}(0)$$

  – Gives the estimation error in terms of the initial error.

- Does this guarantee that $\tilde{\mathbf{x}} = 0 \; \forall \; t$?
  Or even that $\tilde{\mathbf{x}} \to 0$ as $t \to \infty$? (which is a more realistic goal).

  – Response is fine if $\tilde{\mathbf{x}}(0) = 0$. But what if $\tilde{\mathbf{x}}(0) \neq 0$?

- If A stable, then $\tilde{\mathbf{x}} \to 0$ as $t \to \infty$, but the dynamics of the estimation error are completely determined by the open-loop dynamics of the system (eigenvalues of A).
  – Could be very slow.
  – No obvious way to modify the estimation error dynamics.

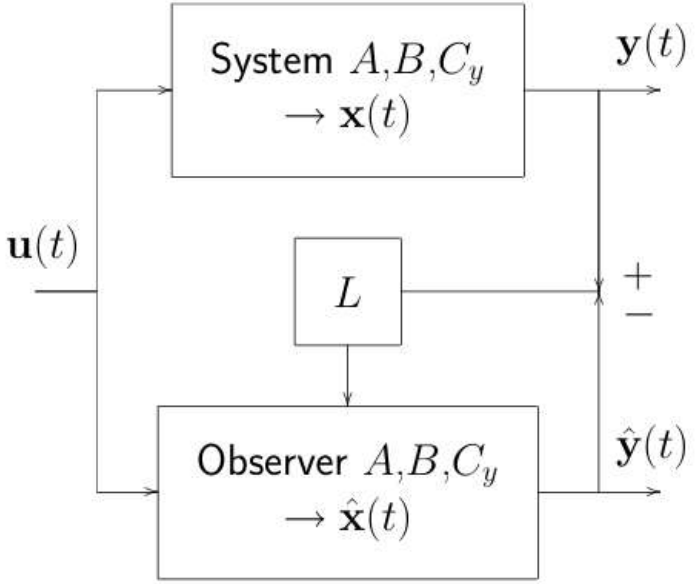
- Open-loop estimation **is not a very good idea.**

Closed-loop Estimator 1

- Obvious fix to problem: use the additional information available:
  - How well does the estimated output match the measured output?

$$\text{Compare: } \mathbf{y} = C_y \mathbf{x} \text{ with } \hat{\mathbf{y}} = C_y \hat{\mathbf{x}}$$

  - Then form $\tilde{\mathbf{y}} = \mathbf{y} - \hat{\mathbf{y}} \equiv C_y \tilde{\mathbf{x}}$



Closed-loop Estimator 2

- **Approach:** Feedback $\tilde{\mathbf{y}}$ to improve our estimate of the state. Basic form of the estimator is:

$$\begin{aligned}\dot{\hat{\mathbf{x}}}(t) &= A\hat{\mathbf{x}}(t) + B\mathbf{u}(t) + \boxed{L\tilde{\mathbf{y}}(t)} \\ \hat{\mathbf{y}}(t) &= C_y\hat{\mathbf{x}}(t)\end{aligned}$$

where L is a **user selectable gain matrix.**

- **Analysis:**

$$\begin{aligned}\dot{\tilde{\mathbf{x}}} = \dot{\mathbf{x}} - \dot{\hat{\mathbf{x}}} &= [A\mathbf{x} + B\mathbf{u}] - [A\hat{\mathbf{x}} + B\mathbf{u} + L(\mathbf{y} - \hat{\mathbf{y}})] \\ &= A(\mathbf{x} - \hat{\mathbf{x}}) - L(C\mathbf{x} - C_y\hat{\mathbf{x}}) \\ &= A\tilde{\mathbf{x}} - LC_y\tilde{\mathbf{x}} = (A - LC_y)\tilde{\mathbf{x}}\end{aligned}$$

Closed-loop Estimator 3

- So the closed-loop estimation error dynamics are now

$$\dot{\tilde{\mathbf{x}}} = (A - LC_y)\tilde{\mathbf{x}} \quad \text{with solution} \quad \tilde{\mathbf{x}}(t) = e^{(A-LC_y)t}\,\tilde{\mathbf{x}}(0)$$

- **Bottom line:** Can select the gain L to attempt to improve the convergence of the estimation error (and/or speed it up).
  – But now must worry about observability of the system $[A, C_y]$.

- Note the similarity:
  – **Regulator Problem:** pick K for $A - BK$
    ◇ Choose $K \in \mathcal{R}^{1\times n}$ (SISO) such that the closed-loop poles

$$\det(sI - A + BK) = \Phi_c(s)$$

    are in the desired locations.
  – **Estimator Problem:** pick L for $A - LC_y$
    ◇ Choose $L \in \mathcal{R}^{n\times 1}$ (SISO) such that the closed-loop poles

$$\det(sI - A + LC_y) = \Phi_o(s)$$

    are in the desired locations.

Example

$$
\begin{aligned}
A &= \begin{bmatrix} -1 & 1.5 \\ 1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad x(0) = \begin{bmatrix} -0.5 \\ -1 \end{bmatrix} \\
C_y &= \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0
\end{aligned}
$$

- Assume that the initial conditions are not well known.
- System stable, but $\lambda_{\max}(A) = -0.18$
- Test observability:

$$
\texttt{rank} \begin{bmatrix} C_y \\ C_y A \end{bmatrix} = \texttt{rank} \begin{bmatrix} 1 & 0 \\ -1 & 1.5 \end{bmatrix}
$$

- Use open and closed-loop estimators. Since the initial conditions are not well known, use $\hat{\mathbf{x}}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

- Open-loop estimator:

$$
\begin{aligned}
\dot{\hat{\mathbf{x}}} &= A\hat{\mathbf{x}} + B\mathbf{u} \\
\hat{\mathbf{y}} &= C_y\hat{\mathbf{x}}
\end{aligned}
$$

Solution 1

- Closed-loop estimator:

$$\begin{aligned}\dot{\hat{\mathbf{x}}} &= A\hat{\mathbf{x}} + B\mathbf{u} + L\tilde{\mathbf{y}} = A\hat{\mathbf{x}} + B\mathbf{u} + L(\mathbf{y} - \hat{\mathbf{y}}) \\ &= (A - LC_y)\hat{\mathbf{x}} + B\mathbf{u} + L\mathbf{y} \\ \hat{\mathbf{y}} &= C_y\hat{\mathbf{x}}\end{aligned}$$

  - Dynamic system with poles $\lambda_i(A - LC_y)$ that takes the measured plant outputs as an input and generates an estimate of $\mathbf{x}$.
  - Use `place` command to set closed-loop pole locations

Solution 2

- Open-loop case:

$$
\begin{aligned}
\dot{\mathbf{x}} &= A\mathbf{x} + B\mathbf{u} \\
\mathbf{y} &= C_y\mathbf{x} \\
\dot{\hat{\mathbf{x}}} &= A\hat{\mathbf{x}} + B\mathbf{u} \\
\hat{\mathbf{y}} &= C_y\hat{\mathbf{x}}
\end{aligned}
$$

$$
\Rightarrow \begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\hat{\mathbf{x}}} \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \hat{\mathbf{x}} \end{bmatrix} + \begin{bmatrix} B \\ B \end{bmatrix} \mathbf{u} \;, \quad \begin{bmatrix} \mathbf{x}(0) \\ \hat{\mathbf{x}}(0) \end{bmatrix} = \begin{bmatrix} -0.5 \\ -1 \\ 0 \\ 0 \end{bmatrix}
$$

$$
\begin{bmatrix} \mathbf{y} \\ \hat{\mathbf{y}} \end{bmatrix} = \begin{bmatrix} C_y & 0 \\ 0 & C_y \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \hat{\mathbf{x}} \end{bmatrix}
$$

Solution 3

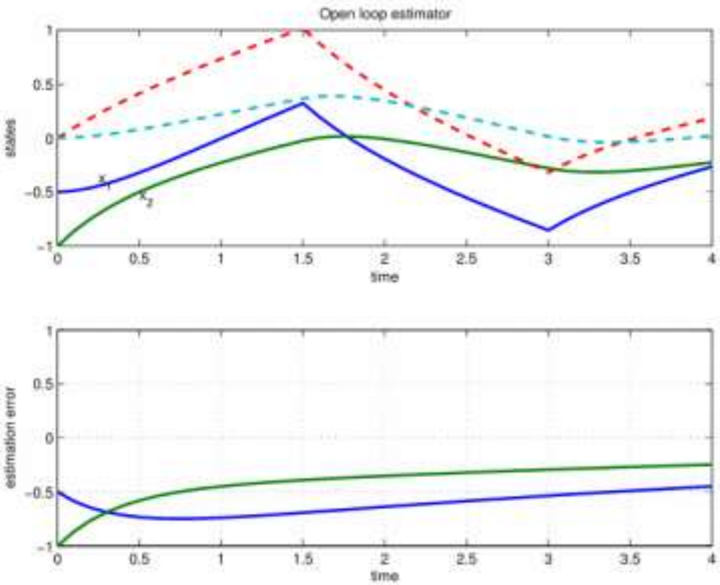
- Closed-loop case:

$$\begin{aligned}\dot{\mathbf{x}} &= A\mathbf{x} + B\mathbf{u} \\ \dot{\hat{\mathbf{x}}} &= (A - LC_y)\hat{\mathbf{x}} + B\mathbf{u} + LC_y\mathbf{x}\end{aligned}$$

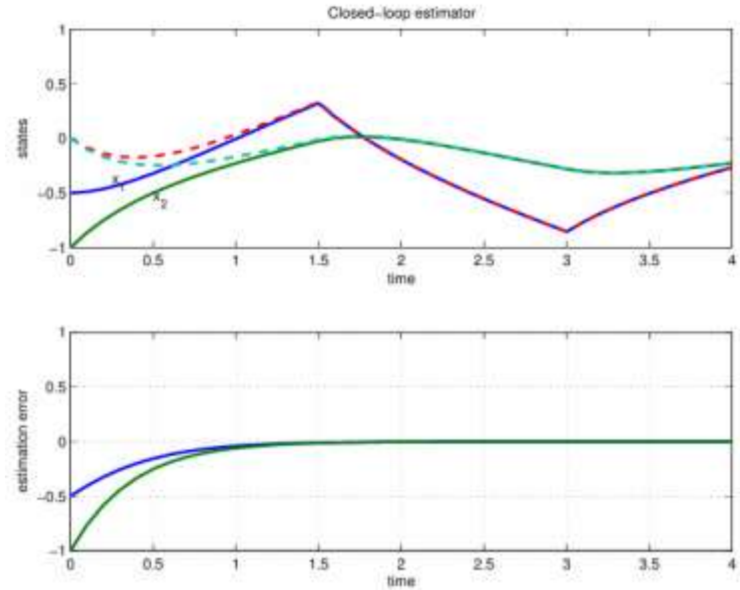
$$\Rightarrow \begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\hat{\mathbf{x}}} \end{bmatrix} = \begin{bmatrix} A & 0 \\ LC_y & A - LC_y \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \hat{\mathbf{x}} \end{bmatrix} + \begin{bmatrix} B \\ B \end{bmatrix} \mathbf{u}$$

- Example uses a strong $\mathbf{u}(t)$ to shake things up

Solution 4



Open-loop estimator. Estimation error



Closed-loop estimator. Convergence looks much better.

Estimator Poles?

- Location heuristics for poles still apply – use Bessel, ITAE, ...
  - Main difference: probably want to make the estimator faster than you intend to make the regulator – should enhance the control, which is based on $\hat{\mathbf{x}}(t)$.

- **Note:** When designing a regulator, were concerned with "bandwidth" of the control getting too high $\Rightarrow$ often results in control commands that *saturate* the actuators and/or change rapidly.

- Different concerns for the estimator:
  - Loop closed inside computer, so saturation not a problem.
  - However, the measurements $\mathbf{y}$ are often "noisy", and we need to be careful how we use them to develop our state estimates.

What shoulb be the next step?

⇒ **High bandwidth estimators** tend to accentuate the effect of sensing noise in the estimate.

– State estimates tend to "track" the measurements, which are fluctuating randomly due to the noise.

⇒ **Low bandwidth estimators** have lower gains and tend to rely more heavily on the plant model

– Essentially an open-loop estimator – tends to ignore the measurements and just uses the plant model.